NSW GOVERNMENT

Never Eat Soggy Australians

2022 HIGHER SCHOOL CERTIFICATE EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A data sheet, formulae sheet, and Periodic Table are provided at the back of this paper

Total marks: 100

Section I – 20 marks (pages 2–12)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 12–29)

- Attempt Questions 21–32
- Allow about 2 hours and 25 minutes for this section

Section I

20 marks
Attempt Questions 1–20
Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

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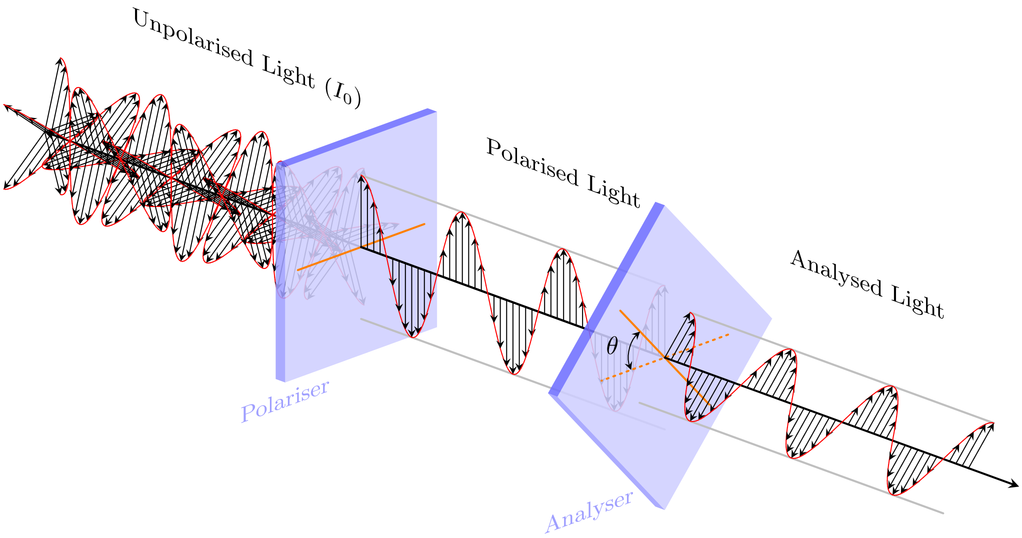
1 A polarising filter is setup as shown, with the magnetic fields depicted. What value for θ will prevent 60 % of the unpolarised light intensity from passing through the analyser?



A. 50.77°

B. 26.57°

C. 39.23°

D. It is not possible.

2 Which of the following eras during the Big Bang are in the correct order?

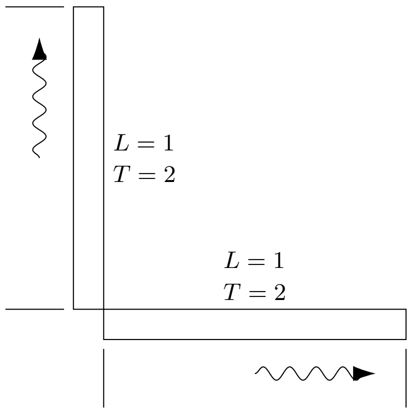
A. Electroweak Era, Era of Nuclei, Grand Unified Theory Era, Era of Atoms

B. Grand Unified Theory Era, Era of Nuclei, Era of Nucleosynthesis, Era of Atoms

C. Planck Era, Particle Era, Era of Nuclei, Era of Atoms,

D. Planck Era, Era of Nucleosynthesis, Particle Era, Era of Atoms

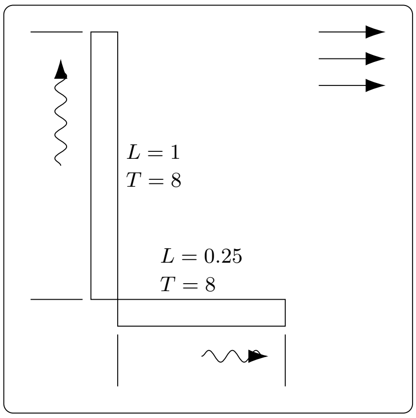
3 A "light clock" is an idealised clock consisting of a rod which is one light second long, with two mirrors at either end, such that light is reflected. One period of the clock is measured as the time it takes to go to one end and be reflected back.



NOT TO SCALE

Given that the light clocks below are moving at a velocity such that $\sqrt{1 - \frac{v^2}{c^2}} = 0.25$, which of the following situations conforms to both of Einstein's postulates when viewed by a stationary observer?

A.



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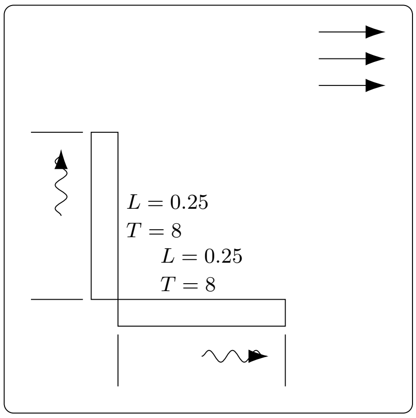

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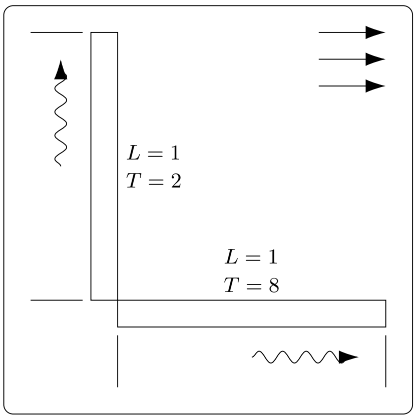
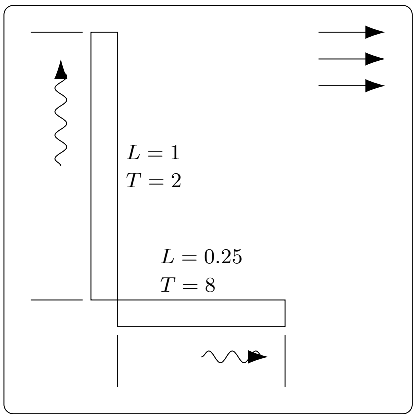


D.



4 When high-energy positive pions collide with protons in bubble chambers, they can transform into positive kaons and sigma-plus baryons.

$$\pi^+ + p \rightarrow \mathrm{K}^+ + \Sigma^+$$

Reactant	Mass (u)	Product	Mass (u)
π^+	0.149833	K^+	0.52998
p	1.00728	Σ^+	1.27683

How much energy is released in the above reaction?

A. $605.205\,\mathrm{MeV}$

B. $0.649710 \times c^2\,\mathrm{J}$

C. $0.649\,710\,\mathrm{u}$

D. $605.205\,\mathrm{MeV/c^2}$

5 The table shows three flavours of quarks and their respective charges and strangeness. Strangeness is conserved in strong interactions.

Quark	Symbol	Charge	Strangeness
Up	u	$+\frac{2}{3}$	0
Down	d	$-\frac{1}{3}$	0
Strange	s	$-\frac{1}{3}$	-1

In a particular nuclear transformation involving only the strong interaction, a meson with quark composition $\bar{u}d$ and a baryon with quark composition uud transform into a meson with quark composition $d\bar{s}$ and a baryon with composition:

A. $ss\bar{s}$

B. udd

C. $\bar{u}\bar{d}\bar{s}$

D. uds

6 π Scorpii is a triple star system located in the Scorpius Constellation, at a distance of 590 ly from Earth.. Its luminosity is roughly 21900 suns and has an effective black body temperature of 24 960 °C. Which of the following is wavelength of peak intensity for this star?

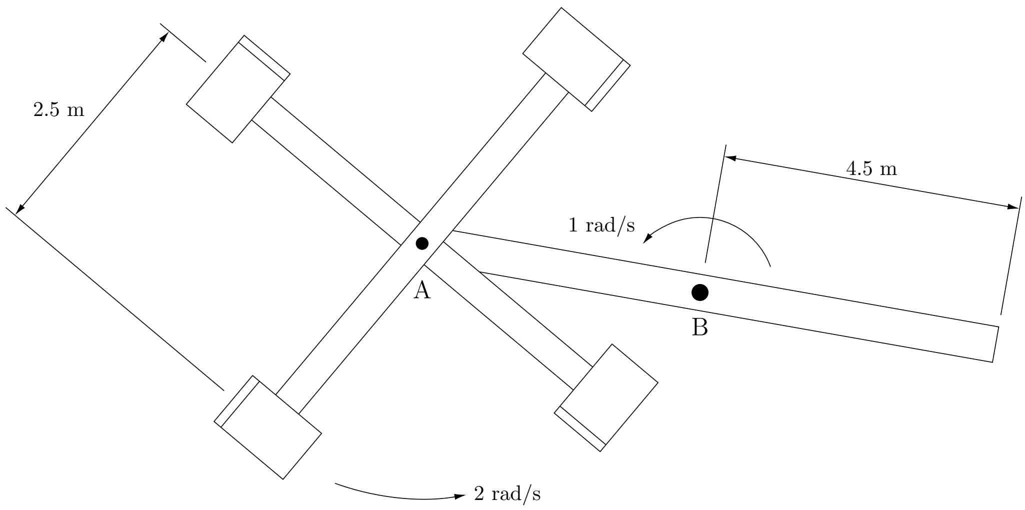
A. 116 nm

B. 115 nm

C. 117 nm

D. 1161 nm

7 The seats on an amusement park ride, shown below, rotate in the plane of the ground. The arm rotates around axle B with a constant angular velocity of $1\,\mathrm{rad\,s^{-1}}$. The minor arm rotates around axle A with a constant angular velocity of $2\,\mathrm{rad\,s^{-1}}$. What is the maximum centripetal force that an 80 kg rider will undergo?



A. 5040 N

B. 1604 N

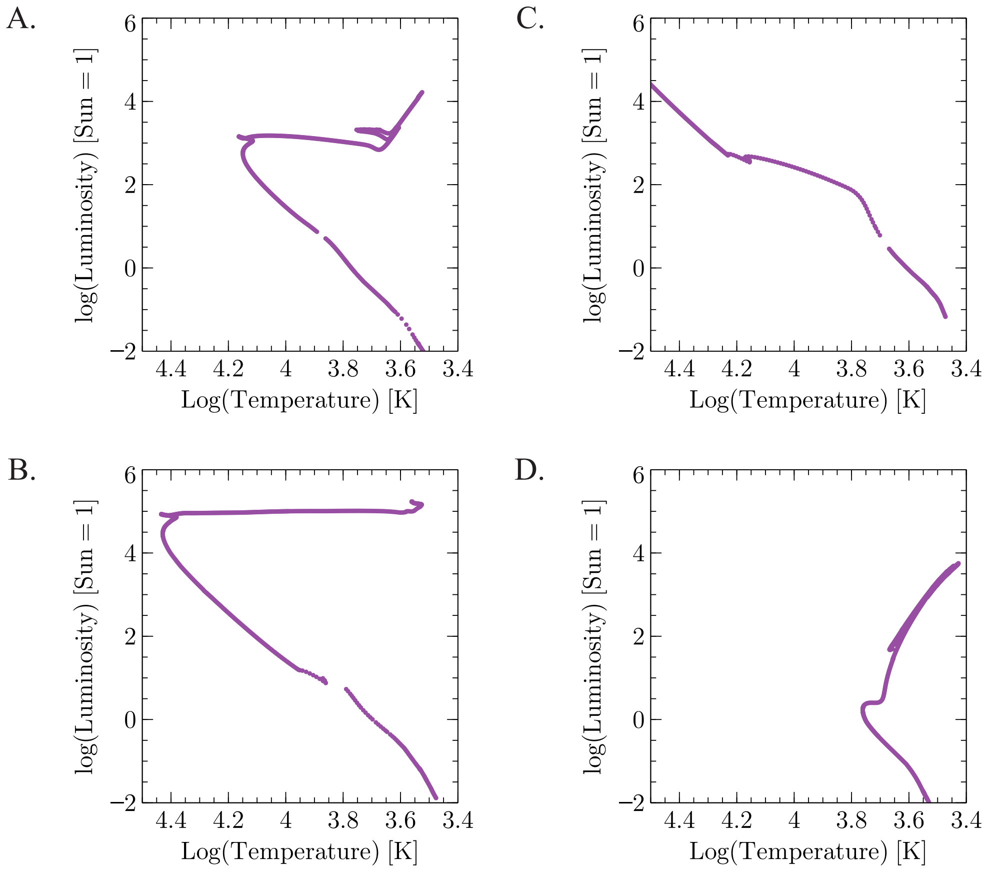
C. 1646 N

D. 1160 N

8 A keen physics student sets up an experiment to demonstrate the photoelectric effect. They shine a light of a certain intensity onto a photo-cathode, and plot the photo-current against the applied voltage. Noting that the light intensity is $I = nhf$, where n is the photon flux and f is the frequency of the light, if the frequency increases, what is the expected change to the stopping voltage and the saturation current?

	Stopping Voltage	*Saturation Current*
A	Increases	The same
B	Increases	Decreases
C	The same	Increases
D	Decreases	The same

9 The Hertzsprung-Russell diagram shows the positions of stars in four clusters of different ages. Which of the following clusters is the oldest?



10 In an experiment similar to Young's double slit, coherent monochromatic light passes through two slits 1 mm apart, forming a diffraction pattern on a distant screen. If the light is of frequency 3.0 THz, how many diffraction minima will appear on the screen?

A. 21

B. 20

C. 10

D. 9

11 A wire 1000 m long has a resistance per unit length of $25\,\mathrm{m\Omega\,m^{-1}}$. This line provides power to a load which draws 10 A at 110 V. By calculating the power loss in the transmission wire, calculate the turn ratio ($n_p : n_s$) in a transformer with $V_p = 3000\,\mathrm{V}$ required to ensure the load receives the correct voltage.

A. 27:1

B. 8:1

C. 1:27

D. 1:12

12 Which of the following expressions gives the orbital velocity of a satellite in terms of the total energy T, assuming the orbit is circular?

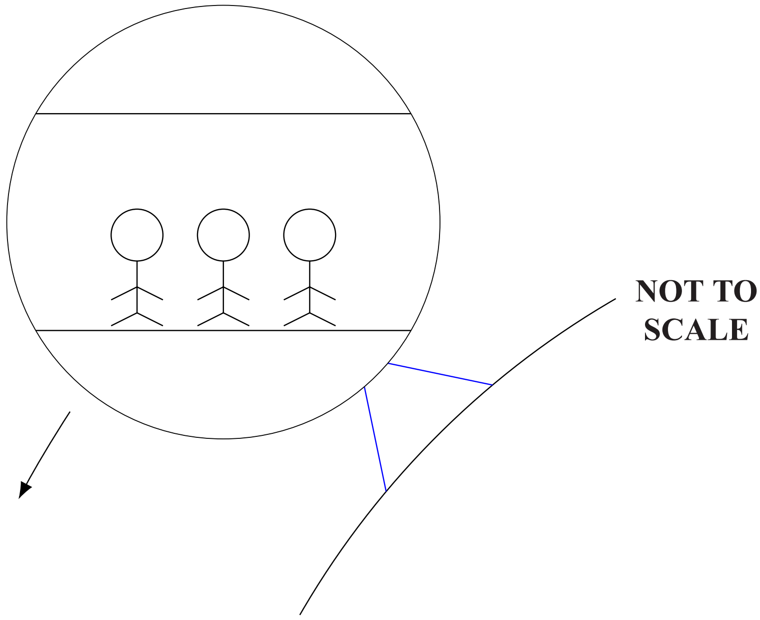
A. $\sqrt{\frac{T}{m}}$

B. $\sqrt{\frac{2T}{m}}$

C. $\sqrt{-\frac{2T}{m}}$

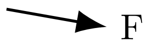
D. $\sqrt{-\frac{T}{2m}}$

13 The diagram below shows a carriage moving around a Ferris Wheel.

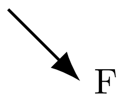


Which alternative below shows the direction of the force exerted *on the carriage by the supports* as the carriage moves around the Ferris Wheel?

A.

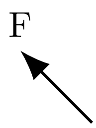


C.



B.

D.



14 What is the energy in electron volts required to ionise the electrons from three hydrogen atoms from the ground state?

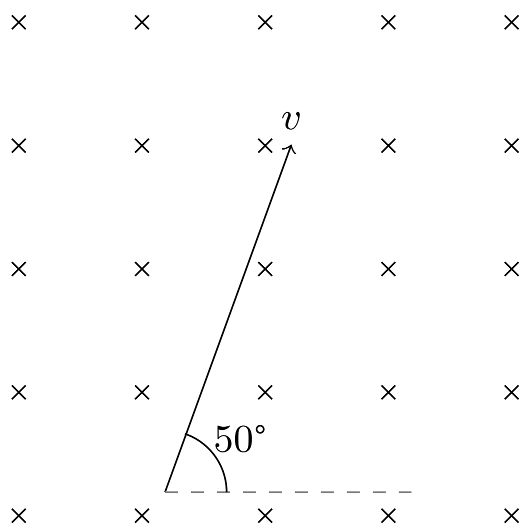
A. 40.8 eV

B. 13.6 eV

C. 11.3 eV

D. 13.6 eV

15 What is the magnitude of the force applied to the electron when it is travelling at a velocity of $7 \times 10^2\,\mathrm{m\,s^{-1}}$ in a magnetic field of strength 5 mT?



A. 5.6×10^{-16} N

B. 3.6×10^{-16} N

C. 4.3×10^{-19} N

D. 5.6×10^{-19} N

16 What action would be performed to rapidly halt the reactions within a nuclear reactor?

A. Removing the fuel rods.

B. Inserting the control rods.

C. Removing the neutron moderator.

D. Removing the control rods.

17 Which of the following provided experimental evidence for De Broglie's matter-waves?

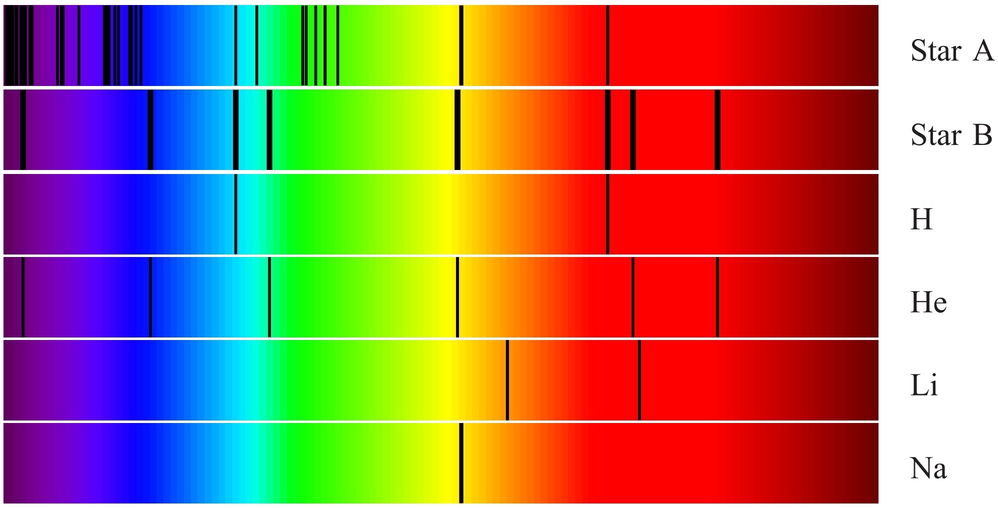
A. Electron diffraction off crystalline structures.

B. The photoelectric effect.

C. Hydrogen spectral lines derived from the quantisation of angular momentum as the radii of electron orbits are integer multiples of the wavelength of the electrons.

D. The divergence between the predicted and observed intensity emitted from a black body at low wavelengths.

18 Which elements are common to both stars, and which star is denser?



	Common Elements	*Denser Star*
A	H, He, Na	Star A
B	H	Star B
C	Li, Na	Star A
D	H, Na	Star B

19 A DC motor consists of a square coil with side lengths $3\,\text{cm}$ and 70 turns within a magnetic field of strength $0.11\,\text{T}$. At $t = 0$ it is in a position such that the magnetic flux through the coil is zero. Assuming that a point of the outermost part of the coil rotates at a constant speed $v = 6\,\text{cm}\,\text{s}^{-1}$, what is the magnitude of the torque produced per ampere at $t = 2\,\text{s}$?

A. $0.65\,\text{N}\,\text{m}\,\text{A}^{-1}$

B. $0.75\,\text{N}\,\text{m}\,\text{A}^{-1}$

C. $4.53\,\text{N}\,\text{mm}\,\text{A}^{-1}$

D. $5.24\,\text{N}\,\text{mm}\,\text{A}^{-1}$

20 What is the current required to produce a force between two parallel conducting wires $\pi\,\text{m}$ long and $2\,\mu\text{m}$ apart equivalent to that required to accelerate an electron at $1.10 \times 10^{30}\,\text{m}\,\text{s}^{-2}$.

A. $1.78\,\text{A}$

B. $1.87\,\text{PA}$

C. $3.16\,\text{A}$

D. $3.18\,\text{A}$

2022 HIGHER SCHOOL CERTIFICATE EXAMINATION

Centre Number

Student Number

Physics

Section II Answer Booklet

80 marks
Attempt Questions 21–32
Allow about 2 hours and 25 minutes for this section

Instructions

- Write your Centre Number and Student Number at the top of this page
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of the response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

Do NOT write in this area.

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Question 21 (6 marks)

Two projectiles are launched at the same time with velocities $40.0\,\mathrm{m\,s^{-1}}$ and $25.0\,\mathrm{m\,s^{-1}}$, at $60.0°$ and $36.9°$, respectively. Projectile (1) is launched from $10.0\,\mathrm{m}$ to the left of Projectile (2). Take $g = 10\,\mathrm{m\,s^{-2}}$.

(a) Determine the maximum heights of both projectiles. **3**

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(b) Determine the distance between the highest points in the particles' journeys. **3**

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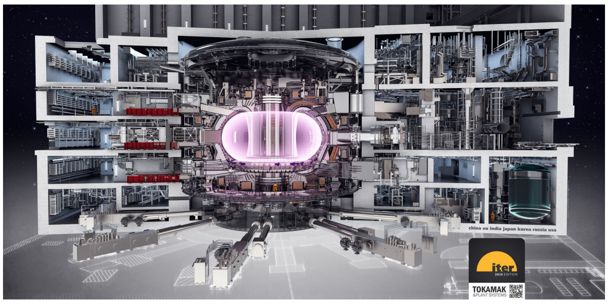
Question 22 (7 marks)

The International Thermonuclear Experimental Reactor (ITER) project is a nuclear fusion research project. It generates an magnetic field using a central solenoid (Figure 1b) which is 18 m tall, has 4050 turns, is 4.3 m wide, and weighs over 1000 tonnes. The magnetic field is used to contain the plasma and heat it up to high temperatures.



(a) An image of the reactor design.

(b) Part of the central solenoid.

ITER produces energy through the fusion of deuterium and tritium (isotopes of hydrogen):

$$\underset{2.014102u}{{}^{2}_{1}\mathrm{D}} + \underset{3.016049u}{{}^{3}_{1}\mathrm{T}} \rightarrow \underset{4.001504u}{{}^{4}_{2}\mathrm{He}} + \underset{1.008665u}{{}^{1}_{0}\mathrm{n}}$$

This fusion reaction can happen above 100 000 000 K and ITER's plasma core will reach temperatures of 150 000 000 K. Neutrons emitted by the reactor are absorbed by the walls of the reactor and could be used to heat water into steam.

(a) When fully operational, 50 MW of heating power is expected to produce 500 MW of output energy. Explain how the operation of ITER remains consistent with the law of conservation of energy. Include a calculation in your answer. **3**

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(b) Calculate the magnetic field strength inside the central solenoid when it operates at 46 kA. **2**

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(c) With reference to appropriate physics principles, explain why neutrons are able to escape the magnetic field and why the plasma can not. **2**

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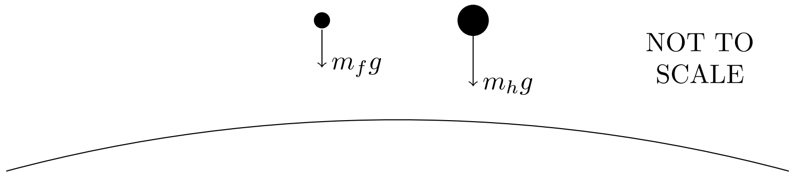
Question 23 (6 marks)

A physics teacher informed their class that a hammer will not fall to the ground faster than a feather, ignoring the effects of air resistance. They explained that because the force (F) exerted on the objects is equal to the product of the mass (m) and the gravitational acceleration (g), the acceleration of both the hammer and the feather is therefore $g = 9.81\,\mathrm{m\,s^{-2}}$, and it takes both objects the same time to fall.



(a) However, a keen physics student noticed that both the earth (of mass m_e) and the falling body (of mass m_b) would exert a force on each other, and thus they would take different times to fall. Calculate the magnitude of the net acceleration of the falling body relative to earth, making sure justify your choice of reference frame. **3**

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(b) One of the physics student's even keener classmates notes that the mass of the hammer (m_h) and the feather (m_f) come from the earth. Explain, with reference to appropriate physics principles, why the teacher is actually correct. **3**

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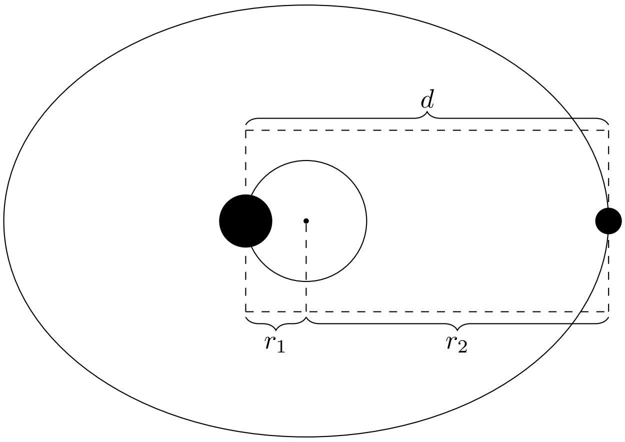
Question 24 (8 marks)

Consider the following solar system within a fictional universe:



(a) Given that the larger planet is twice the mass of the smaller planet, find the radius of each planet's orbit around a fixed point in terms of the distance d between the two planets. **3**

Hint: The planets orbit about their centre of mass.

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(b) Hence find the ratio of the orbital periods, making sure to carefully justify your reasoning. **2**

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(c) If $m_1 = 2.17 \times 10^{10}$ tonne, $r_1 = 3.00 \times 10^8$ mm and $T_2 = 10.0$ years, find the gravitational constant G in this universe. **3**

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Question 25 (4 marks)

A keen physics student decides to sit on the hood of a car that is tipping into a river on a rock embankment. The average student weighs 60.0 kg, the mass of the car is 1.40 tonne, the centre of mass is 70.0 cm behind the front axle, the wheelbase is 2.3 m long, the rock embankment contacts the bottom of the car 46.0 cm behind the front axle and the wheels have a diameter of 0.00 m.

(a) Considering that it has a fulcrum at the point where the embankment contacts the bottom of the car, calculate the torque the car produces. **2**

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(b) Calculate the number of friends the student needs sitting on the hood which extends 40.0 cm forward from the front axle to stop it from tipping in to the river. **2**

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Question 26 (7 marks)

(a) X rays with a wavelength of 71 pm are directed onto a gold foil and eject tightly bound electrons from the gold atoms. The ejected electrons them move in circular paths of a radius r in a region of a uniform magnetic field B. For the fastest of the ejected electrons, the product Br is equal to 1.88×10^{-4} T m. Find the maximum kinetic energy of those electrons and the work done in removing them from the gold atoms. Ignore any relativistic effects. **4**

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(b) If the de Broglie wavelength of a proton is 100 fm, find the electric potential needed to accelerate a proton to this speed. **3**

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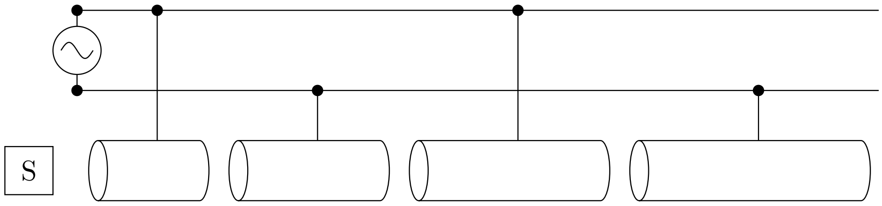
Question 27 (8 marks)

(a) Explain how particle accelerator(s) have obtained evidence that investigates SPECIFIC aspects of the Standard Model of Matter and ONE other theory. **5**

(b) In 2050, Scientists at the Contemporary CERN (*ConCERN*) design a LINAC to accelerate protons, which are produced by a source S. The LINAC consists of drift tubes of increasing length, which are connected to an AC source of 100 kV at 200 MHz such that the ends of each adjacent cylinder have opposite polarities at the that which the particles are synched to leave the tube. **3**



Calculate the kinetic energy of the protons in MeV at the 100th tube, and how long this tube must be.

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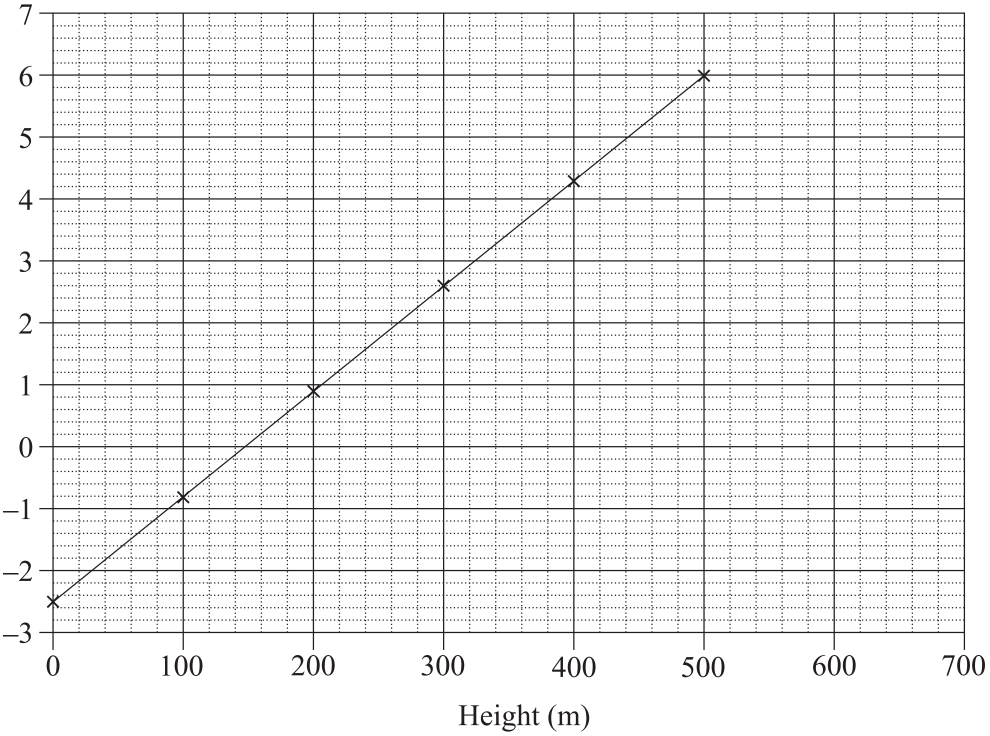
Question 28 (8 marks)

Muons are subatomic particles produced in the upper atmosphere with a velocity of $0.996c$. They exponentially decay according to the equation $N_t = N_0 e^{-t/t_0}$. A keen physics wished to replicate Rossi and Hall's 1941 experiment in more depth. They set up six muon detectors, 100 m vertically apart, and measured the muon fluxes over a one hour period in each.

Height (m)	Muon Flux		
500	400		
400	351		
300	302		
200	249		
100	222		
0	184		

(a) Write a linear relationship between the height of the detector and the relative muon fluxes and use this to fill in the blank columns. Then, plot the results on the graph below. **3**



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(b) Using part (a), calculate the mean lifetime (t_0) of the muons in their reference frame. **2**

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(c) The graph in (a) also shows the classically-predicted results. Explain how Rossi-Hall's experiment provides evidence for the Theory of Special Relativity. **3**

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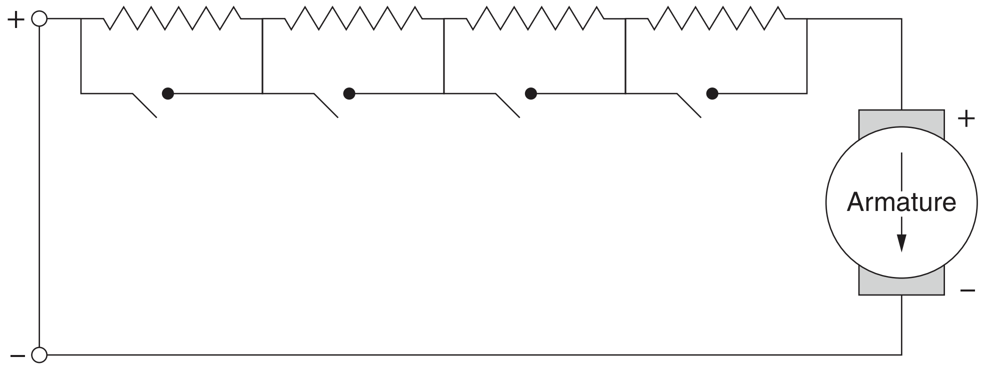
Question 29 (6 marks)

This diagram shows an electric circuit used to control a DC electric motor.



(a) Explain the operation of a DC motor with reference to physical principles, and how the electric circuit can be used to control the speed of rotation. **4**

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(b) Describe how the speed of rotation in an AC motor is controlled. **2**

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Question 30 (5 marks)

(a) Describe the problem caused by the homogeneity of the Cosmic Microwave Background (CMBR), and how it was resolved. **3**

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(b) A few minutes after the Big Bang, the universe has cooled enough that no new matters (protons, neutrons, electrons) are formed from energy. When the temperature of the universe reaches 1×10^9 K, the neutron to proton ratio $\frac{n}{p} \approx 4:25$, and the fusion of particles begins in a series of reactions: **2**

$$^{1}_{1}\mathrm{H} + ^{1}_{0}\mathrm{n} \rightleftharpoons ^{2}_{1}\mathrm{H} + \gamma \qquad (1)$$

$$^{2}_{1}\mathrm{H} + ^{2}_{1}\mathrm{H} \rightleftharpoons ^{3}_{1}\mathrm{H} + ^{1}_{1}\mathrm{H} \qquad (2)$$

$$^{3}_{1}\mathrm{H} + ^{2}_{1}\mathrm{H} \rightleftharpoons ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n} \qquad (3)$$

By writing the net reaction, explain why the (mass) ratio of hydrogen to helium in the universe is roughly 3:1.

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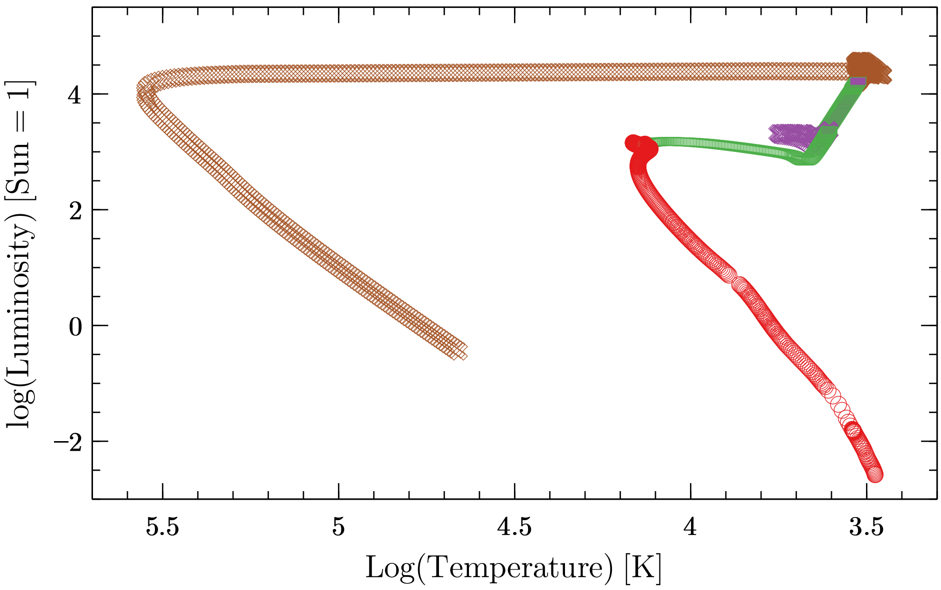
Question 31 (8 marks)



The Hertzsprung-Russell Diagram above shows the evolution of stars within a globular cluster. Label the evolutionary stages shown on the diagram, and explain the changes to the key features of the spectra of a star of mass $5M_{\odot}$ as it evolves.

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Question 31 (continued)

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Question 32 (7 marks)

(a) Analyse ONE experiment which determined the speed of light and relate it to Maxwell's prediction of the velocity of an electromagnetic wave. **4**

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(b) Explain how the photoelectric effect differs from the classical model of light in TWO ways. **3**

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Acknowledgements

This exam was produced by modelmat#8218. This specific version was produced on August 25, 2022. This work, except for the parts listed below, is licensed under a Creative Commons Attribution–NonCommercial 4.0 International License. DO NOT redistribute without this page. Explicitly, this paper is NOT for use, adaptation, or distribution by tutoring centres.

The diagram for Question 1 is modified from `https://tex.stackexchange.com/a/114029/`.

The idea for Question 3 is from `https://sites.pitt.edu/~jdnorton/teaching/HPS_0410/chapters/Special_relativity_clocks_rods/index.html`.

Question 7 is from 1999 Engineering Science 3U, Question 3(a).

Question 9 and 31 uses data from Harvard's MESA Isochrones & Stellar Tracks project, and the diagrams were helpfully produced by Mr S.

Images for Question 22 are Credit © ITER Organisation, `https://www.iter.org`.

The idea for Question 23 came from `https://physics.stackexchange.com/q/3534/`.

Question 25 was a contribution from L.D.

Question 26 is from Halliday's Fundamentals of Physics (2013), Module 38 Problems 21 and 59.

Question 29 is an extension of 2020 Engineering Studies, Question 26(d), with the diagram stolen because I couldn't be bothered learning how to write circuitikz.

Question 31 is a crime against humanity.

Version Log

- 19/11/21: Initial Release
- 20/11/21: Adjusted answers for Question 19, as mistakes were made.